

Are size and mitochondrial power of cells inter-determined?

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Mitochondria are the central hub of ATP production in most eukaryotic cells. Cellular power (energy per unit time), which is primarily generated in these organelles, is crucial to our understanding of cell function in health and disease. We investigated the relation between a mitochondrion's power (metabolic rate) and host cell size by combining metabolic theory with the analysis of two recent databases, one covering 109 protists and the other 63 species including protists, metazoans, microalgae, and vascular plants. We uncovered an interesting statistical regularity: in well-fed protists, relatively elevated values of mitochondrion power cluster around the smallest cell sizes and the medium-large cell sizes. In contrast, in starved protists and metazoans, the relation between mitochondrion power and cell size is inconclusive, and in microalgae and plants, mitochondrion power seems to increase from smaller cells to larger ones (where this investigation includes plant cells of volume up to ca. $2.18 \times 10^5 \mu\text{m}^3$). Using these results, estimates are provided of the number of active ATP synthase molecules and basal uncouplers.

Approaches to monitor ATP levels in living cells: where do we stand?

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ATP is the most universal and essential energy molecule in cells. This is due to its ability to store cellular energy in form of high-energy phosphate bonds, which are extremely stable and readily usable by the cell. This energy is key for a variety of biological functions such as cell growth and division, metabolism, and signaling, and for the turnover of biomolecules. Understanding how ATP is produced and hydrolyzed with a spatiotemporal resolution is necessary to understand its functions both in physiological and in pathological contexts. In this review, first we will describe the organization of the electron transport chain and ATP synthase, the main molecular motor for ATP production in mitochondria. Second, we will review the biochemical assays currently available to estimate ATP quantities in cells, and we will compare their readouts, strengths, and weaknesses. Finally, we will explore the palette of genetically encoded biosensors designed for microscopy-based approaches, and show how their spatiotemporal resolution opened up the possibility to follow ATP levels in living cells.

The Mitochondrial Response to DNA Damage.

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Mitochondria are double membrane organelles in eukaryotic cells that provide energy by generating adenosine triphosphate (ATP) through oxidative phosphorylation. They are crucial to many aspects of cellular metabolism. Mitochondria contain their own DNA that encodes for essential proteins involved in the execution of normal mitochondrial functions. Compared with nuclear DNA, the mitochondrial DNA (mtDNA) is more prone to be affected by DNA damaging agents, and accumulated DNA damages may cause mitochondrial dysfunction and drive the pathogenesis of a variety of human diseases, including neurodegenerative disorders and cancer. Therefore, understanding better how mtDNA damages are repaired will facilitate developing therapeutic strategies. In this review, we focus on our current understanding of the mtDNA repair system. We also discuss other mitochondrial events promoted by

excessive DNA damages and inefficient DNA repair, such as mitochondrial fusion, fission, and mitophagy, which serve as quality control events for clearing damaged mtDNA.

Smart Design of Mitochondria-Targeted and ROS-Responsive CPI-613 Delivery Nanoplatfrom for Bioenergetic Pancreatic Cancer Therapy.

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Mitochondria, as the powerhouse of most cells, are not only responsible for the generation of adenosine triphosphate (ATP) but also play a decisive role in the regulation of apoptotic cell death, especially of cancer cells. Safe potential delivery systems which can achieve organelle-targeted therapy are urgently required. In this study, for effective pancreatic cancer therapy, a novel mitochondria-targeted and ROS-triggered drug delivery nanoplatfrom was developed from the TPP-TK-CPI-613 (TTCI) prodrug, in which the ROS-cleave thioketal functions as a linker connecting mitochondrial targeting ligand TPP and anti-mitochondrial metabolism agent CPI-613. DSPE-PEG2000 was added as an assistant component to increase accumulation in the tumor via the EPR effect. This new nanoplatfrom showed effective mitochondrial targeting, ROS-cleaving capability, and robust therapeutic performances. With active mitochondrial targeting, the formulated nanoparticles (TTCI NPs) demonstrate much higher accumulation in mitochondria, facilitating the targeted delivery of CPI-613 to its acting site. The results of in vitro antitumor activity and cell apoptosis revealed that the IC₅₀ values of TTCI NPs in three types of pancreatic cancer cells were around 20~30 μ M, which was far lower than those of CPI-613 (200 μ M); 50 μ M TTCI NPs showed an increase in apoptosis of up to 97.3% in BxPC3 cells. Therefore, this mitochondria-targeted prodrug nanoparticle platform provides a potential strategy for developing safe, targeting and efficient drug delivery systems for pancreatic cancer therapy.

Beyond ATP, new roles of mitochondria.

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Mitochondria, special double-membraned intracellular compartments or 'organelles', are popularly known as the 'powerhouses of the cell', as they generate the bulk of ATP used to fuel cellular biochemical reactions. Mitochondria are also well known for generating metabolites for the synthesis of macromolecules (e.g., carbohydrates, proteins, lipids and nucleic acids). In the mid-1990s, new evidence suggesting that mitochondria, beyond their canonical roles in bioenergetics and biosynthesis, can act as signalling organelles began to emerge, bringing a dramatic shift in our view of mitochondria's roles in controlling cell function. Over the next two and half decades, works from multiple groups have demonstrated how mitochondrial signalling can dictate diverse physiological and pathophysiological outcomes. In this article, we will briefly discuss different mechanisms by which mitochondria can communicate with cytosol and other organelles to regulate cell fate and function and exert paracrine effects. Our molecular understanding of mitochondrial communication with the rest of the cell, i.e. mitochondrial signalling, could reveal new therapeutic strategies to improve health and ameliorate diseases.

Adrenal Mitochondria and Steroidogenesis: From Individual Proteins to Functional Protein Assemblies.

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The adrenal cortex is critical for physiological function as the central site of glucocorticoid and mineralocorticoid synthesis. It possesses a great degree of specialized compartmentalization at multiple hierarchical levels, ranging from the tissue down to the molecular levels. In this paper, we discuss this functionalization, beginning with the tissue zonation of the adrenal cortex and how this impacts steroidogenic output. We then discuss the cellular biology of steroidogenesis, placing special emphasis on the mitochondria. Mitochondria are classically known as the "powerhouses of the cell" for their central role in respiratory adenosine triphosphate synthesis, and attention is given to mitochondrial electron transport, in both the context of mitochondrial respiration and mitochondrial steroid metabolism. Building on work demonstrating functional assembly of large protein complexes in respiration, we further review research demonstrating a role for multimeric protein complexes in mitochondrial cholesterol transport, steroidogenesis, and mitochondria-endoplasmic reticulum contact. We aim to highlight with this review the shift in steroidogenic cell biology from a focus on the actions of individual proteins in isolation to the actions of protein assemblies working together to execute cellular functions.

Mitochondrial transplantation and transfer: The promising method for diseases.

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Mitochondria are organelles that serve as the powerhouses for cellular bioenergetics in eukaryotic cells. It is responsible for mitochondrial adenosine triphosphate (ATP) generation, cell signaling and activity, calcium balance, cell survival, proliferation, apoptosis, and autophagy. Mitochondrial transplantation is a promising disease therapy that involves the recovery of mitochondrial dysfunction using isolated functioning mitochondria. The objective of the present article is to provide current knowledge on natural mitochondrial transfer processes, in vitro and in vivo applications of mitochondrial transplantation, clinical trials, and challenges associated with mitochondrial transplantation.

Regulation of proteostasis and innate immunity via mitochondria-nuclear communication.

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Mitochondria are perhaps best known as the "powerhouse of the cell" for their role in ATP production required for numerous cellular activities. Mitochondria have emerged as an important signaling organelle. Here, we first focus on signaling pathways mediated by mitochondria-nuclear communication that promote protein homeostasis (proteostasis). We examine the mitochondrial unfolded protein response (UPR^m) in *C. elegans*, which is regulated by a transcription factor harboring both a mitochondrial- and nuclear-targeting sequence, the integrated stress response in mammals, as well as the regulation of chromatin by mitochondrial metabolites. In the second section, we explore the role of mitochondria-to-nuclear communication in the regulation of innate immunity and inflammation. Perhaps related to their prokaryotic origin, mitochondria harbor molecules also found in viruses and bacteria. If these molecules accumulate in the cytosol, they elicit the same innate immune responses as viral or

bacterial infection.

Mitochondria as intracellular signalling organelles. An update.

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Traditionally, mitochondria are known as "the powerhouse of the cell," responsible for energy (ATP) generation (by the electron transport chain, oxidative phosphorylation, the tricarboxylic acid cycle, and fatty acid β -oxidation), and for the regulation of several metabolic processes, including redox homeostasis, calcium signalling, and cellular apoptosis. The extensive studies conducted in the last decades portray mitochondria as multifaceted signalling organelles that ultimately command cells' survival or death. Based on current knowledge, we'll outline the mitochondrial signalling to other intracellular compartments in homeostasis and pathology-related mitochondrial stress conditions here. The following topics are discussed: (i) oxidative stress and mtROS signalling in mitohormesis, (ii) mitochondrial Ca^{2+} signalling; (iii) the anterograde (nucleus-to-mitochondria) and retrograde (mitochondria-to-nucleus) signal transduction, (iv) the mtDNA role in immunity and inflammation, (v) the induction of mitophagy- and apoptosis - signalling cascades, (vi) the mitochondrial dysfunctions (mitochondriopathies) in cardiovascular, neurodegenerative, and malignant diseases. The novel insights into molecular mechanisms of mitochondria-mediated signalling can explain mitochondria adaptation to metabolic and environmental stresses to achieve cell survival.

Introduction. Oxidative stress in mitochondria disorders of aging.

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These special issues of Biological Signals and Receptors are intended to describe mitochondrial DNA damage, oxidative stress and human diseases, including neurodegenerative and neuromuscular diseases, disorders associated with aging, and ischemia-perfusion injury. Traditionally, mitochondria have been viewed as the 'powerhouse' of the cell, i.e., the site of the oxidative phosphorylation machinery involved in adenosine triphosphate (ATP) production. Consequently, much of the research conducted on mitochondria over the past 4 decades has focused on elucidating both those molecular events involved in ATP synthesis by oxidative phosphorylation and those involved in the biogenesis of the oxidative phosphorylation machinery. While monumental achievements have been made, and continue to be made, in the study of these remarkable but extremely complex processes essential for the life of most animal cells, it has been only in recent years that a large body of biological and biomedical scientists have come to recognize that mitochondria participate in other important processes. Two of these are cell death and aging which, not surprisingly, are related processes both involving, in part, the oxidative phosphorylation machinery. This new awareness has sparked a new and growing area of mitochondrial research that has become of great interest to a wide variety of scientists ranging from those involved in elucidating the role of mitochondria in cell death and aging to those interested in either suppressing or facilitating these processes as it relates to identifying new therapies or drugs for human disease.